

2c. Content of Computational thinking, algorithms and programming (J276/02)

This component incorporates and builds on the knowledge and understanding gained in Component 01, encouraging learners to apply this knowledge and understanding using computational thinking. Learners will be introduced to algorithms and programming, learning about programming techniques, how to produce robust programs, computational logic,

translators and facilities of computing languages and data representation. Learners will become familiar with computing related mathematics.

Learners may draw on some of this content when completing the Programming Project.

2.1 Algorithms

Learners should have studied the following:

- computational thinking:
 - abstraction
 - decomposition
 - algorithmic thinking
- standard searching algorithms:
 - binary search
 - linear search
- standard sorting algorithms:
 - bubble sort
 - merge sort
 - insertion sort
- how to produce algorithms using:
 - pseudocode
 - using flow diagrams
- interpret, correct or complete algorithms.